SETK 4643

Thermodynamics

Chapter 11: Refrigeration Cycles

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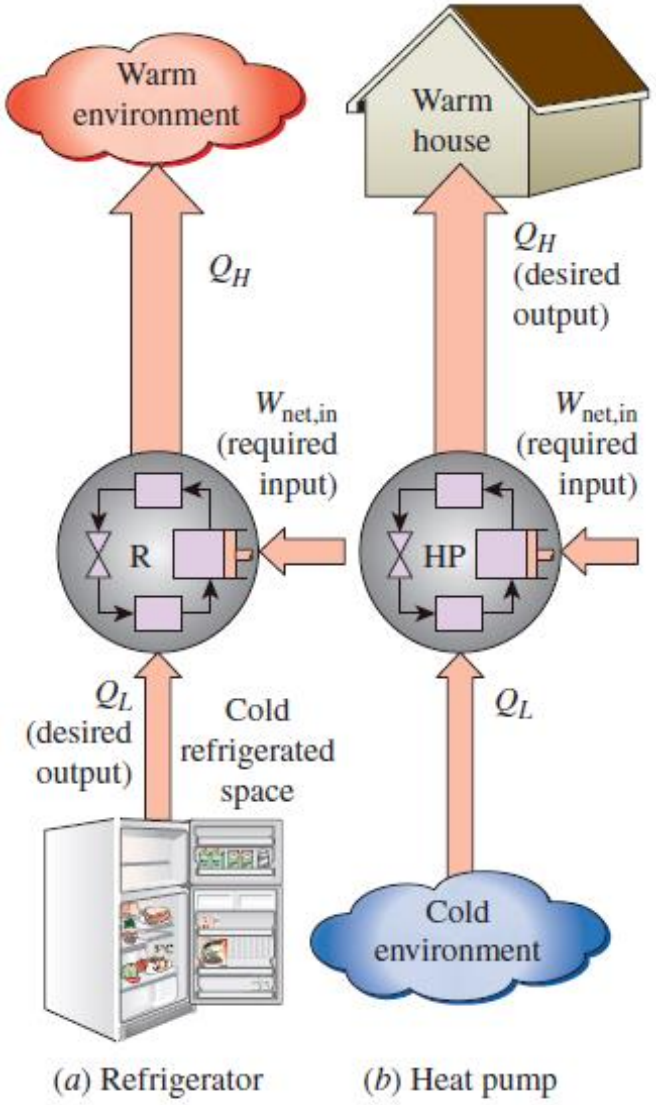


FIGURE 11–1
The objective of a refrigerator is to remove heat (Q_L) from the cold medium; the objective of a heat pump is to supply heat (Q_H) to a warm medium.

11.1 Refrigerators And Heat Pumps

- The transfer of heat from a low-temperature region to a high-temperature one requires special devices called refrigerators.
- Another device that transfers heat from a low-temperature medium to a high-temperature one is the heat pump.
- Refrigerators and heat pumps are essentially the same devices; they differ in their objectives only.

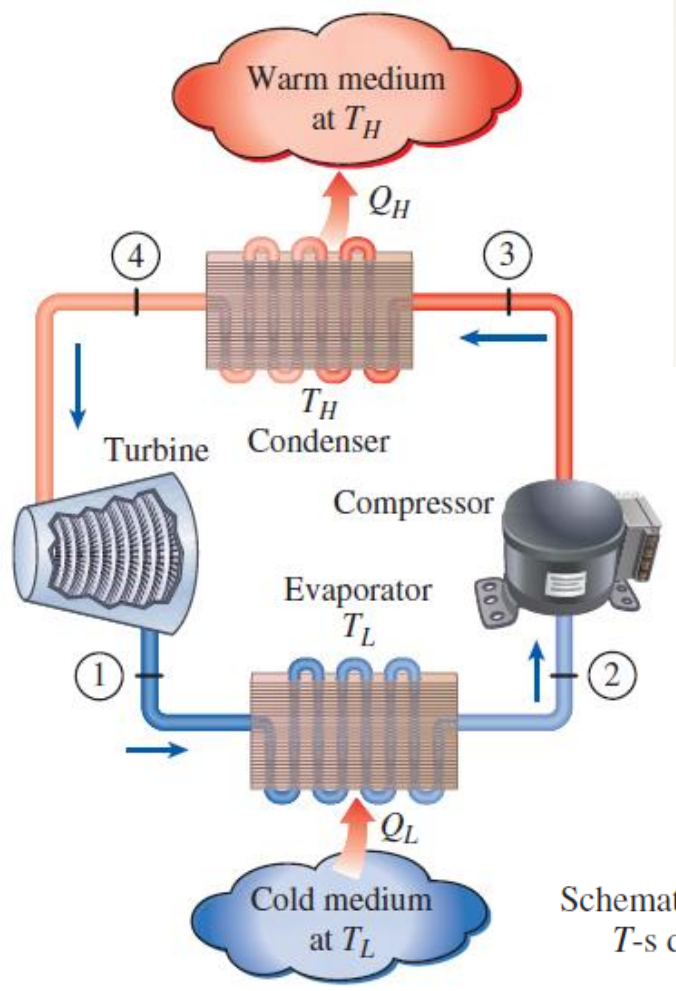
$$\mathrm{COP_R} = \frac{\text{Desired output}}{\text{Required input}} = \frac{\text{Cooling effect}}{\text{Work input}} = \frac{Q_L}{W_{\text{net,in}}}$$

$$\mathrm{COP_{HP}} = \frac{\text{Desired output}}{\text{Required input}} = \frac{\text{Heating effect}}{\text{Work input}} = \frac{Q_H}{W_{\text{net,in}}}$$

$\mathrm{COP_{HP}} = \mathrm{COP_R} + 1$ for fixed values of Q_L and Q_H

11.2 The Reversed Carnot Cycle

- The reversed Carnot cycle is the most efficient refrigeration cycle operating between T_L and T_H.



- It is not a suitable model for refrigeration cycles since processes 2-3 and 4-1 are not practical because.
  - Process 2-3 involves the compression of a liquid–vapor mixture, which requires a compressor that will handle two phases, and
  - process 4-1 involves the expansion of high-moisture-content refrigerant in a turbine.

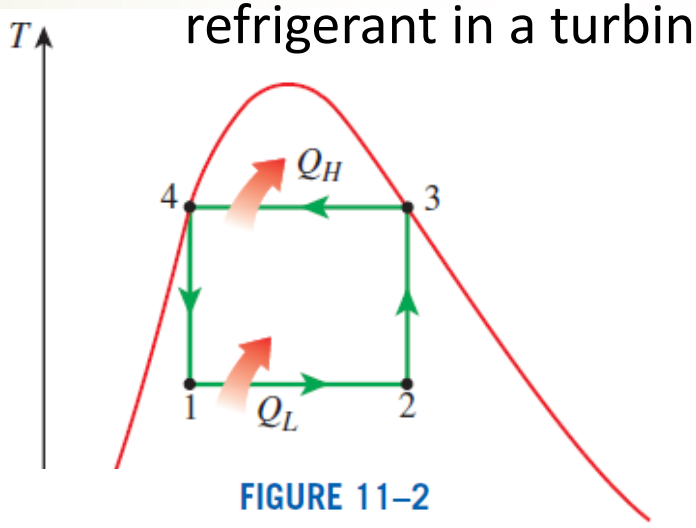


FIGURE 11–2
Schematic of a Carnot refrigerator and T-s diagram of the reversed Carnot cycle.

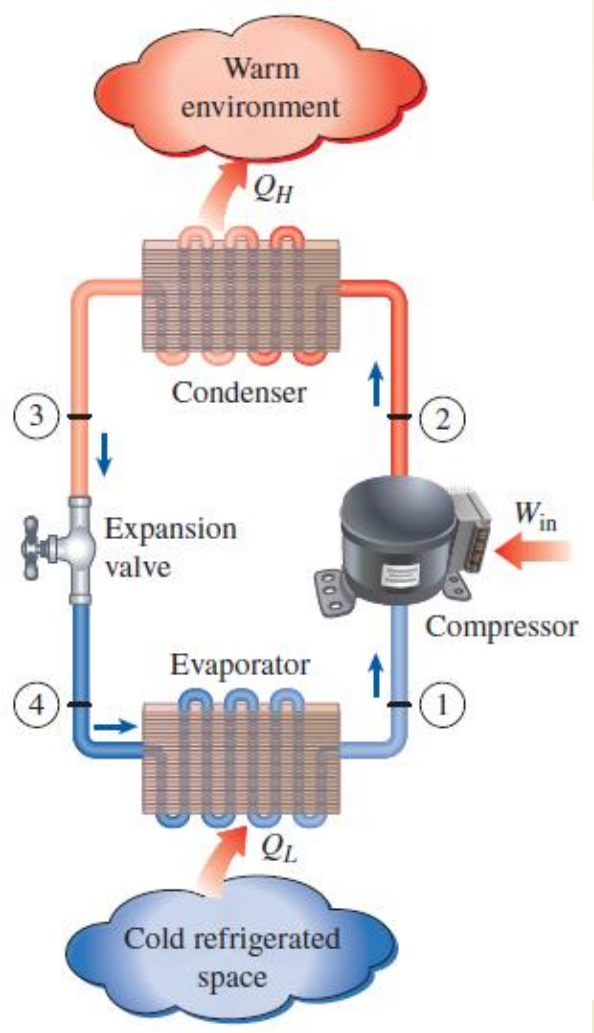
$$\mathrm{COP}_{\mathrm{R,Carnot}} = \frac{1}{T_H/T_L - 1}$$

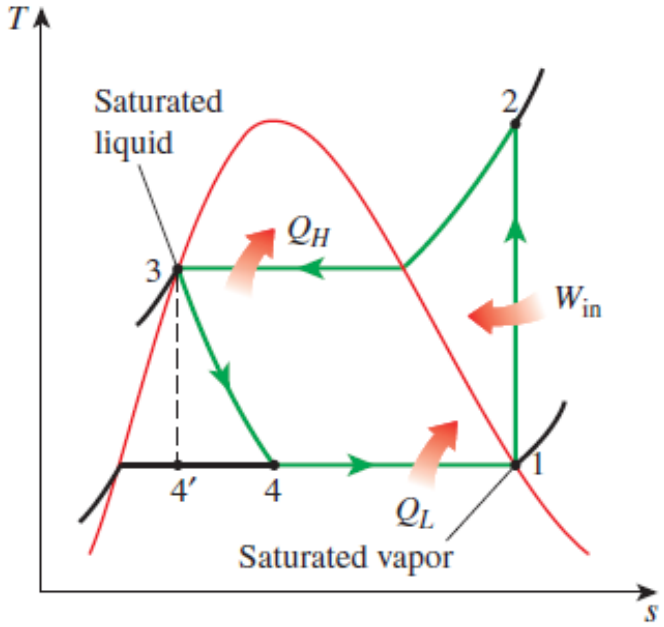
$$\mathrm{COP}_{\mathrm{HP,Carnot}} = \frac{1}{1 - T_L/T_H}$$

- Both COPs increase as the difference between the two temperatures decreases, that is, as T_L rises or T_H falls.

11.3 The Ideal Vapor-compression Refrigeration Cycle

- The vapor-compression refrigeration cycle is the ideal model for refrigeration systems.
  - Unlike the reversed Carnot cycle, the refrigerant is vaporized completely before it is compressed and the turbine is replaced with a throttling device.





Schematic and *T-s* diagram for the ideal vapor-compression refrigeration cycle.

1-2	Isentropic compression in a compressor
2-3	Constant-pressure heat rejection in a condenser
3-4	Throttling in an expansion device
4-1	Constant-pressure heat absorption in an evaporator

- This is the most widely used cycle for refrigerators, A-C systems, and heat pumps.

11.3 The Ideal Vapor-compression Refrigeration Cycle (2)

- The ideal vapor-compression refrigeration cycle involves an irreversible (throttling) process to make it a more realistic model for the actual systems.

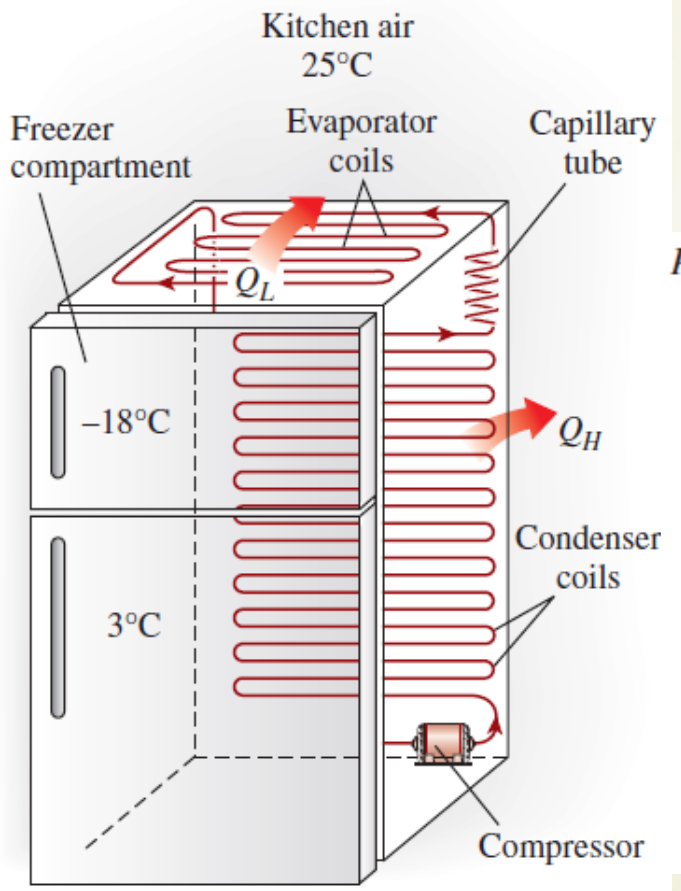


FIGURE 11–4
An ordinary household refrigerator.

- Replacing the expansion valve by a turbine is not practical since the added benefits cannot justify the added cost and complexity.

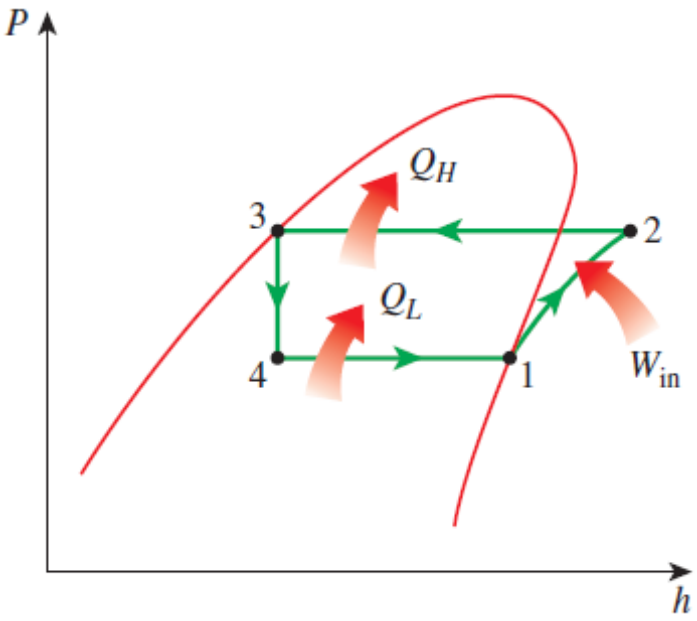


FIGURE 11–5
The *P-h* diagram of an ideal vapor-compression refrigeration cycle.

Steady-flow energy balance:

$$(q_{in} - q_{out}) + (w_{in} - w_{out}) = h_e - h_i$$

$$COP_R = \frac{q_L}{w_{net,in}} = \frac{h_1 - h_4}{h_2 - h_1}$$

$$COP_{HP} = \frac{q_H}{w_{net,in}} = \frac{h_2 - h_3}{h_2 - h_1}$$

$h_1 = h_{g\,@\,P_1}$ and $h_3 = h_{f\,@\,P_3}$ for the ideal case

Example 1

EXAMPLE 11–1 **The Ideal Vapor-Compression Refrigeration Cycle**

A refrigerator uses refrigerant-134a as the working fluid and operates on an ideal vapor-compression refrigeration cycle between 0.14 and 0.8 MPa. If the mass flow rate of the refrigerant is 0.05 kg/s, determine (*a*) the rate of heat removal from the refrigerated space and the power input to the compressor, (*b*) the rate of heat rejection to the environment, and (*c*) the COP of the refrigerator.

SOLUTION A refrigerator operates on an ideal vapor-compression refrigeration cycle between two specified pressure limits. The rate of refrigeration, the power input, the rate of heat rejection, and the COP are to be determined.

Assumptions **1** Steady operating conditions exist. **2** Kinetic and potential energy changes are negligible.

Analysis The *T-s* diagram of the refrigeration cycle is shown in Fig. 11–6. We note that this is an ideal vapor-compression refrigeration cycle, and thus the compressor is isentropic and the refrigerant leaves the condenser as a saturated liquid and enters the compressor as saturated vapor. From the refrigerant-134a tables, the enthalpies of the refrigerant at all four states are determined as follows:

$$P_1 = 0.14 \text{ MPa} \longrightarrow h_1 = h_{g\ @\ 0.14\ \text{MPa}} = 239.19 \text{ kJ/kg}$$
$$s_1 = s_{g\ @\ 0.14\ \text{MPa}} = 0.94467 \text{ kJ/kg·K}$$

$$\left.\begin{array}{l} P_2 = 0.8 \text{ MPa} \\ s_2 = s_1 \end{array}\right\} h_2 = 275.40 \text{ kJ/kg}$$

$$P_3 = 0.8 \text{ MPa} \longrightarrow h_3 = h_{f\ @\ 0.8\ \text{MPa}} = 95.48 \text{ kJ/kg}$$
$$h_4 \cong h_3 \text{ (throttling)} \longrightarrow h_4 = 95.48 \text{ kJ/kg}$$

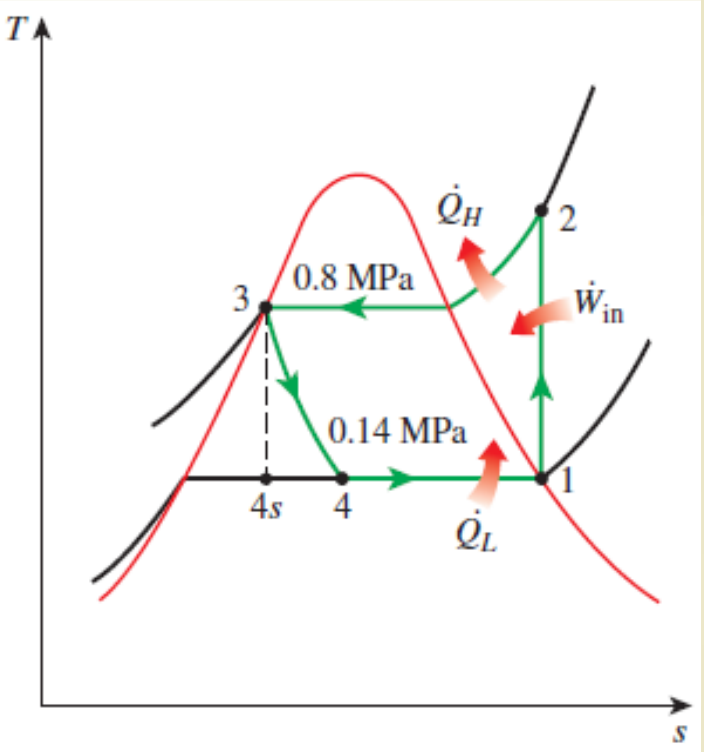


FIGURE 11–6
T-s diagram of the ideal vapor-compression refrigeration cycle described in Example 11–1.

Example 1

(*a*) The rate of heat removal from the refrigerated space and the power input to the compressor are determined from their definitions:

$$\dot{Q}_L = \dot{m}(h_1 - h_4) = (0.05\ \text{kg/s})[(239.19 - 95.48)\ \text{kJ/kg}] = \mathbf{7.19\ kW}$$

and

$$\dot{W}_{in} = \dot{m}(h_2 - h_1) = (0.05\ \text{kg/s})[(275.40 - 239.19)\ \text{kJ/kg}] = \mathbf{1.81\ kW}$$

(*b*) The rate of heat rejection from the refrigerant to the environment is

$$\dot{Q}_H = \dot{m}(h_2 - h_3) = (0.05\ \text{kg/s})[(275.40 - 95.48)\ \text{kJ/kg}] = \mathbf{9.00\ kW}$$

It could also be determined from

$$\dot{Q}_H = \dot{Q}_L + \dot{W}_{in} = 7.19 + 1.81 = 9.00\ \text{kW}$$

(*c*) The coefficient of performance of the refrigerator is

$$\text{COP}_R = \frac{\dot{Q}_L}{\dot{W}_{in}} = \frac{7.19\ \text{kW}}{1.81\ \text{kW}} = \mathbf{3.97}$$

That is, this refrigerator removes about 4 units of thermal energy from the refrigerated space for each unit of electric energy it consumes.

Discussion It would be interesting to see what happens if the throttling valve were replaced by an isentropic turbine. The enthalpy at state 4*s* (the turbine exit with $P_{4s} = 0.14$ MPa, and $s_{4s} = s_3 = 0.35408$ kJ/kg·K) is 88.95 kJ/kg, and the turbine would produce 0.33 kW of power. This would decrease the power input to the refrigerator from 1.81 to 1.48 kW and increase the rate of heat removal from the refrigerated space from 7.19 to 7.51 kW. As a result, the COP of the refrigerator would increase from 3.97 to 5.07, an increase of 28 percent.

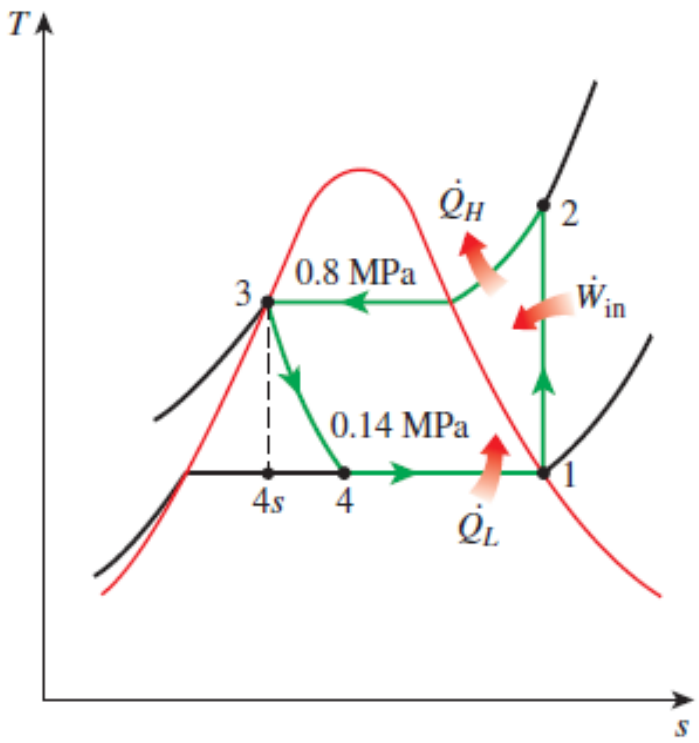
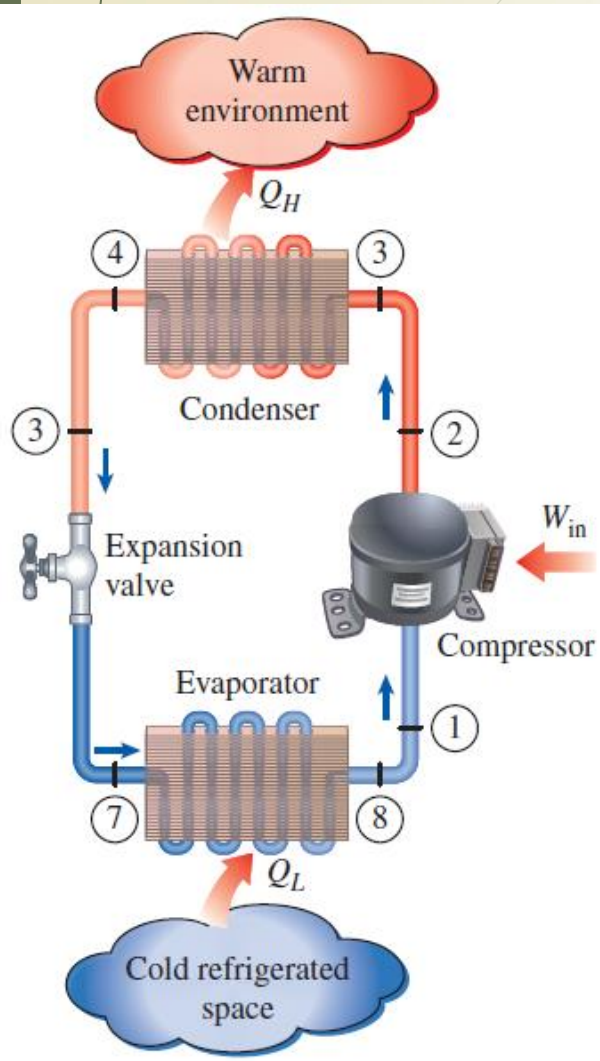


FIGURE 11–6
T-s diagram of the ideal vapor-compression refrigeration cycle described in Example 11–1.

11.4 Actual Vapor-compression Refrigeration Cycle

- An actual vapor-compression refrigeration cycle differs from the ideal one owing mostly to the irreversibilities that occur in various components, mainly due to fluid friction (causes pressure drops) and heat transfer to or from the surroundings.



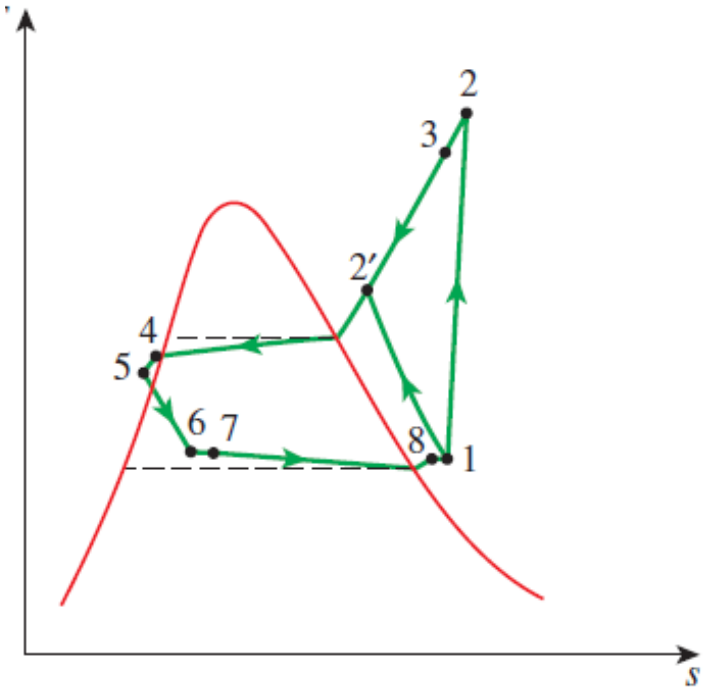


FIGURE 11–7
Schematic and T-s diagram for the actual vapor-compression refrigeration cycle.

- Main different to ideal:
  - Superheated vapor at evaporator exit
  - Non-isentropic compression
  - Subcooled liquid at condenser exit
  - Pressure drops in condenser and evaporator
- The COP decreases as a result of irreversibilities.

Example 2

EXAMPLE 11–2 **The Actual Vapor-Compression Refrigeration Cycle**

Refrigerant-134a enters the compressor of a refrigerator as superheated vapor at 0.14 MPa and −10°C at a rate of 0.05 kg/s and leaves at 0.8 MPa and 50°C. The refrigerant is cooled in the condenser to 26°C and 0.72 MPa and is throttled to 0.15 MPa. Disregarding any heat transfer and pressure drops in the connecting lines between the components, determine (*a*) the rate of heat removal from the refrigerated space and the power input to the compressor, (*b*) the isentropic efficiency of the compressor, and (*c*) the coefficient of performance of the refrigerator.

SOLUTION A refrigerator operating on a vapor-compression cycle is considered. The rate of refrigeration, the power input, the compressor efficiency, and the COP are to be determined.

Assumptions **1** Steady operating conditions exist. **2** Kinetic and potential energy changes are negligible.

Analysis The *T-s* diagram of the refrigeration cycle is shown in Fig. 11–8. We note that the refrigerant leaves the condenser as a compressed liquid and enters the compressor as superheated vapor. The enthalpies of the refrigerant at various states are determined from the refrigerant tables to be

$$\left.\begin{aligned} P_1 &= 0.14 \text{ MPa} \\ T_1 &= -10°\text{C} \end{aligned}\right\} h_1 = 246.37 \text{ kJ/kg}$$

$$\left.\begin{aligned} P_2 &= 0.8 \text{ MPa} \\ T_2 &= 50°\text{C} \end{aligned}\right\} h_2 = 286.71 \text{ kJ/kg}$$

$$\left.\begin{aligned} P_3 &= 0.72 \text{ MPa} \\ T_3 &= 26°\text{C} \end{aligned}\right\} h_3 \cong h_{f@\,26°\text{C}} = 87.83 \text{ kJ/kg}$$

$$h_4 \cong h_3 \text{ (throttling)} \quad \rightarrow \quad h_4 = 87.83 \text{ kJ/kg}$$

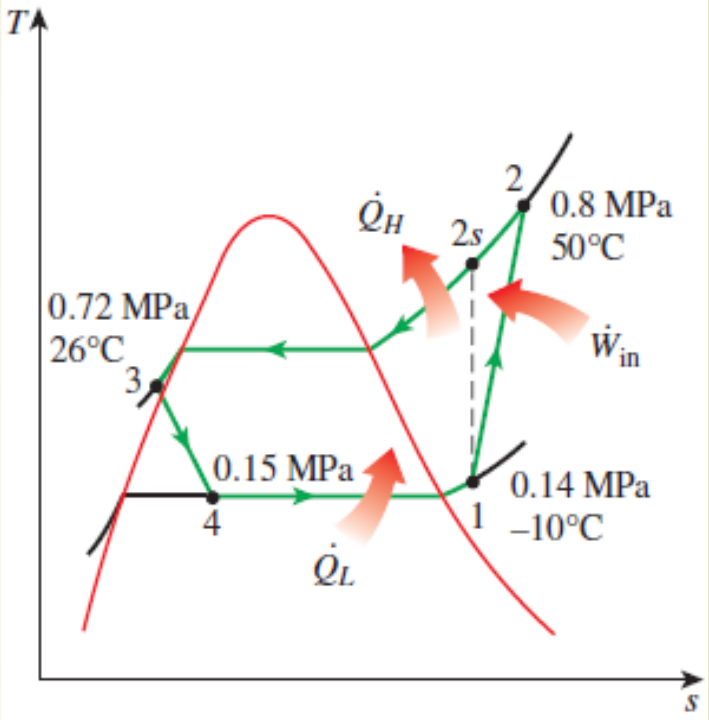


FIGURE 11–8
T-s diagram for Example 11–2.

(*a*) The rate of heat removal from the refrigerated space and the power input to the compressor are determined from their definitions:

$$\dot{Q}_L = \dot{m}(h_1 - h_4) = (0.05\ \text{kg/s})[(246.37 - 87.83)\ \text{kJ/kg}] = \mathbf{7.93\ kW}$$

and

$$\dot{W}_{in} = \dot{m}(h_2 - h_1) = (0.05\ \text{kg/s})[(286.71 - 246.37)\ \text{kJ/kg}] = \mathbf{2.02\ kW}$$

(*b*) The isentropic efficiency of the compressor is determined from

$$\eta_C \cong \frac{h_{2s} - h_1}{h_2 - h_1}$$

where the enthalpy at state 2*s* (P_{2s} = 0.8 MPa and $s_{2s} = s_1$ = 0.9724 kJ/kg·K) is 284.20 kJ/kg. Thus,

$$\eta_C = \frac{284.20 - 246.37}{286.71 - 246.37} = 0.938 \text{ or } \mathbf{93.8\%}$$

(*c*) The coefficient of performance of the refrigerator is

$$\text{COP}_R = \frac{\dot{Q}_L}{\dot{W}_{in}} = \frac{7.93\ \text{kW}}{2.02\ \text{kW}} = \mathbf{3.93}$$

Discussion This problem is identical to the one worked out in Example 11–1, except that the refrigerant is slightly superheated at the compressor inlet and subcooled at the condenser exit. Also, the compressor is not isentropic. As a result, the heat removal rate from the refrigerated space increases (by 10.3 percent), but the power input to the compressor increases even more (by 11.6 percent). Consequently, the COP of the refrigerator decreases from 3.97 to 3.93.

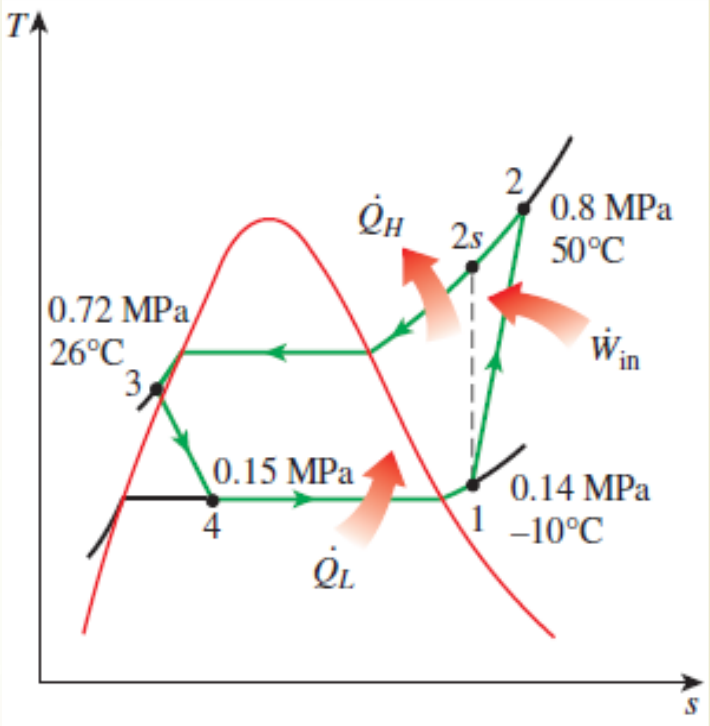


FIGURE 11–8
T-s diagram for Example 11–2.

11.6 Selecting The Right Refrigerant

- Several refrigerants may be used in refrigeration systems such as chlorofluorocarbons (CFCs), ammonia, hydrocarbons (propane, ethane, ethylene, etc.), carbon dioxide, air (in the air-conditioning of aircraft), and even water (above freezing point).
- R-11, R-12, R-22, R-134a, and R-502 account for over 90 percent of the market.
- The industrial and heavy-commercial sectors use ammonia (it is toxic).
- R-11 is used in large-capacity water chillers serving A-C systems in buildings.
- R-134a (replaced R-12, which damages ozone layer) is used in domestic refrigerators and freezers, as well as automotive air conditioners.
- R-22 is used in window air conditioners, heat pumps, air conditioners of commercial buildings, and large industrial refrigeration systems, and offers strong competition to ammonia.

Reading assignment

11.6 Selecting The Right Refrigerant (2)

- R-502 (a blend of R-115 and R-22) is the dominant refrigerant used in commercial refrigeration systems such as those in supermarkets.
- CFCs allow more ultraviolet radiation into the earth's atmosphere by destroying the protective ozone layer and thus contributing to the greenhouse effect that causes global warming.
  - Fully halogenated CFCs (such as R-11, R-12, and R-115) do the most damage to the ozone layer.
  - Refrigerants that are friendly to the ozone layer have been developed.
- Two important parameters that need to be considered in the selection of a refrigerant are the temperatures of the two media (the refrigerated space and the environment) with which the refrigerant exchanges heat.

Reading assignment

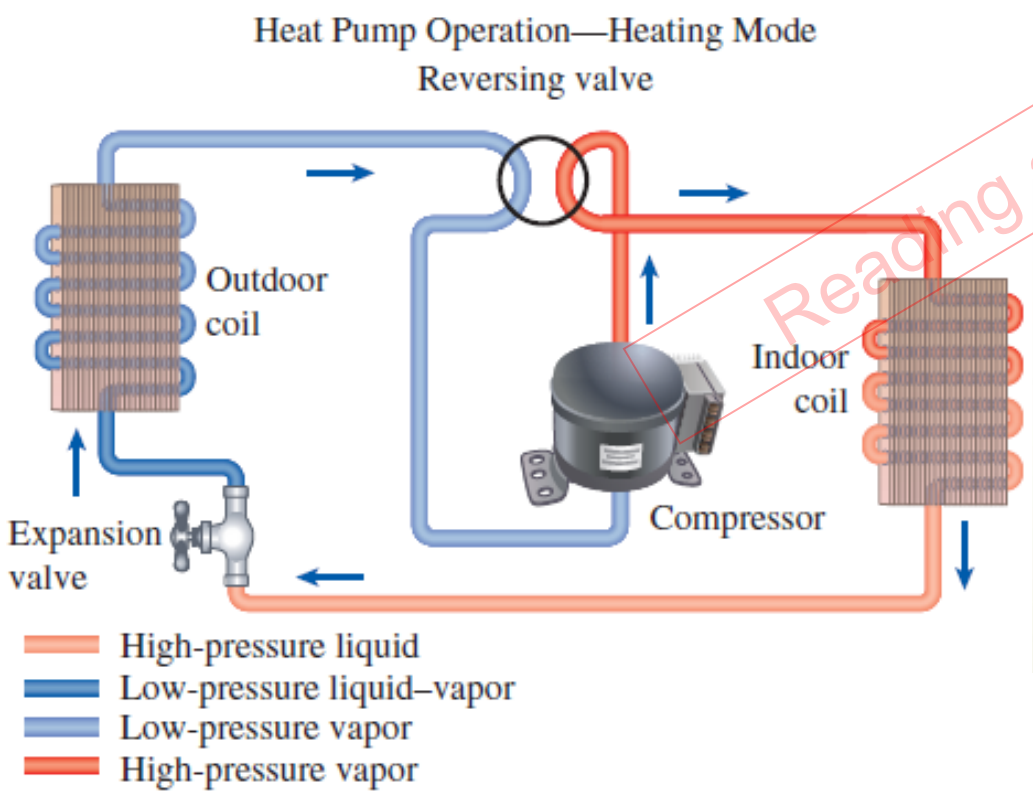
11.7 Heat Pump Systems

- The most common energy source for heat pumps is atmospheric air (air-to-air systems).
- Water-source systems usually use well water and ground-source (geothermal) heat pumps use earth as the energy source.
  - They typically have higher COPs but are more complex and more expensive to install.
- Both the capacity and the efficiency of a heat pump fall significantly at low temperatures.
  - Therefore, most air-source heat pumps require a supplementary heating system such as electric resistance heaters or a gas furnace.

Reading assignment

11.7 Heat Pump Systems (2)

- Heat pumps are most competitive in areas that have a large cooling load during the cooling season and a relatively small heating load during the heating season.
  - In these areas, the heat pump can meet the entire cooling and heating needs of residential or commercial buildings.



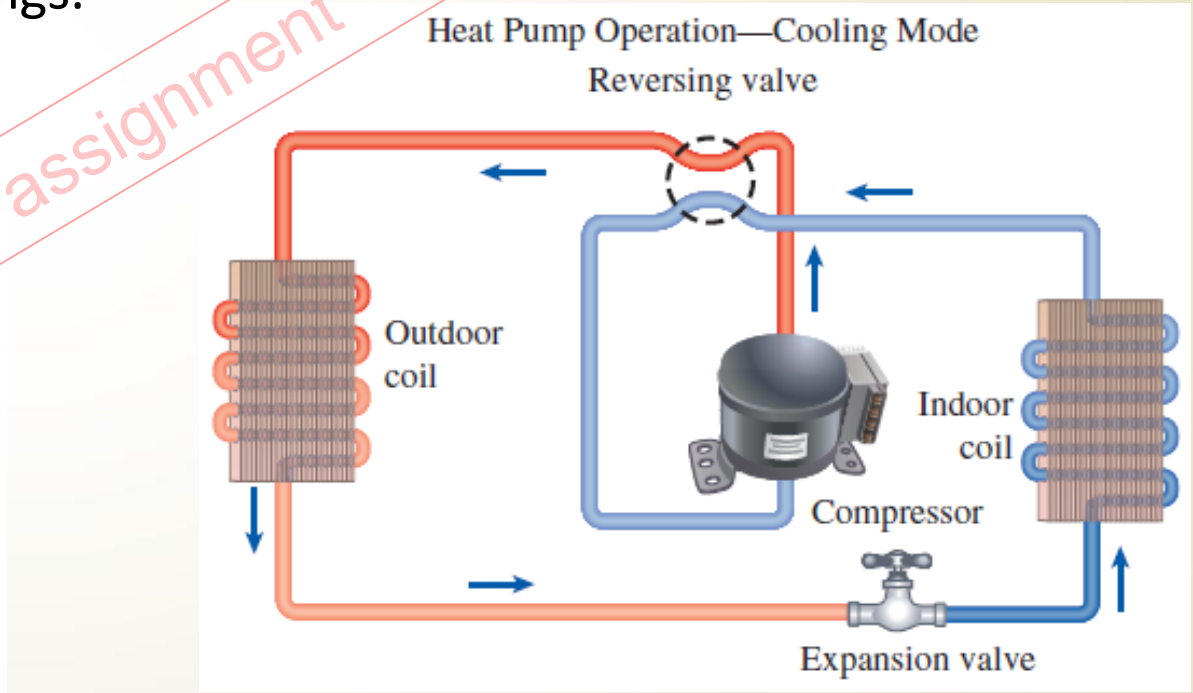
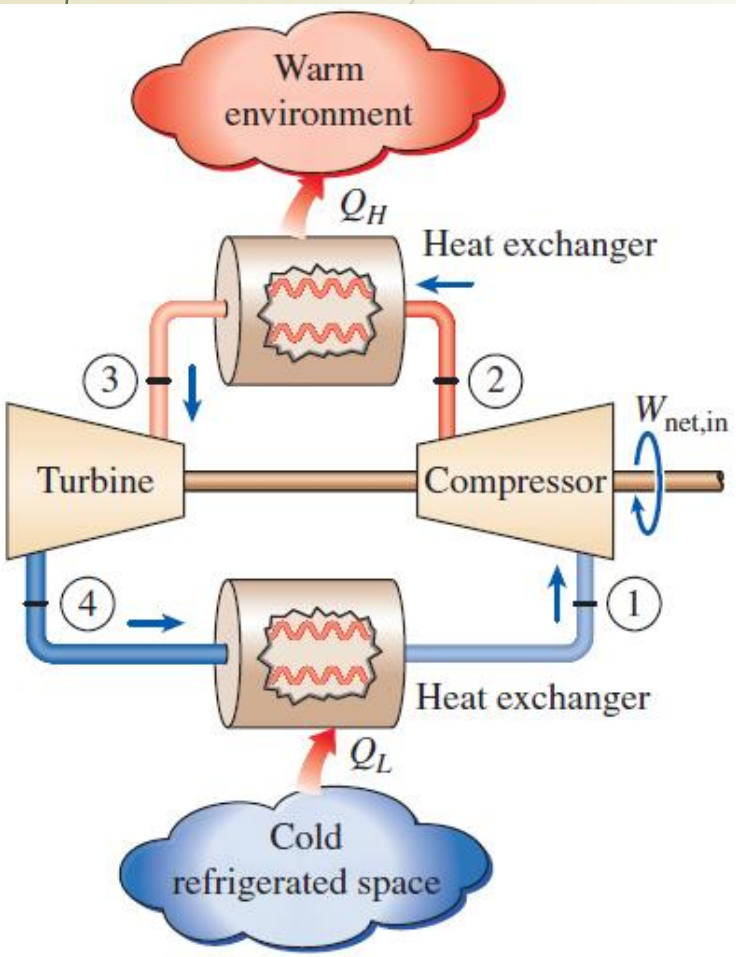


FIGURE 11–11
A heat pump can be used to heat a house in winter and to cool it in summer.

Reading assignment

11.9 Gas Refrigeration Cycles

- The reversed Brayton cycle (the gas refrigeration cycle) can be used for refrigeration.



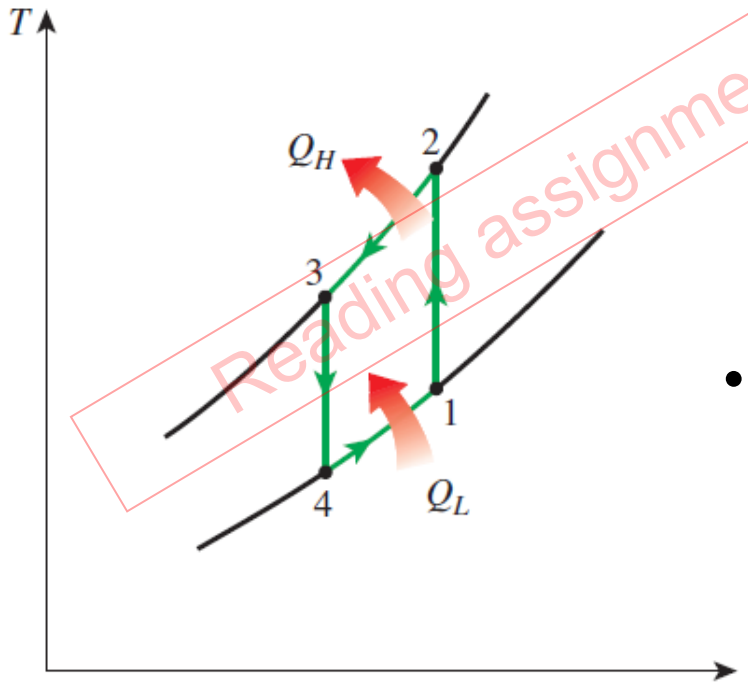


FIGURE 11–18
Simple gas refrigeration cycle.

Reading assignment

$$q_L = h_1 - h_4$$
$$w_{turb,out} = h_3 - h_4$$
$$w_{comp,in} = h_2 - h_1$$

$$COP_R = \frac{q_L}{w_{net,in}} = \frac{q_L}{w_{comp,in} - w_{turb,out}}$$

- The gas refrigeration cycles have lower COPs relative to the vapor-compression refrigeration cycles or the reversed Carnot cycle.

11.9 Gas Refrigeration Cycles (2)

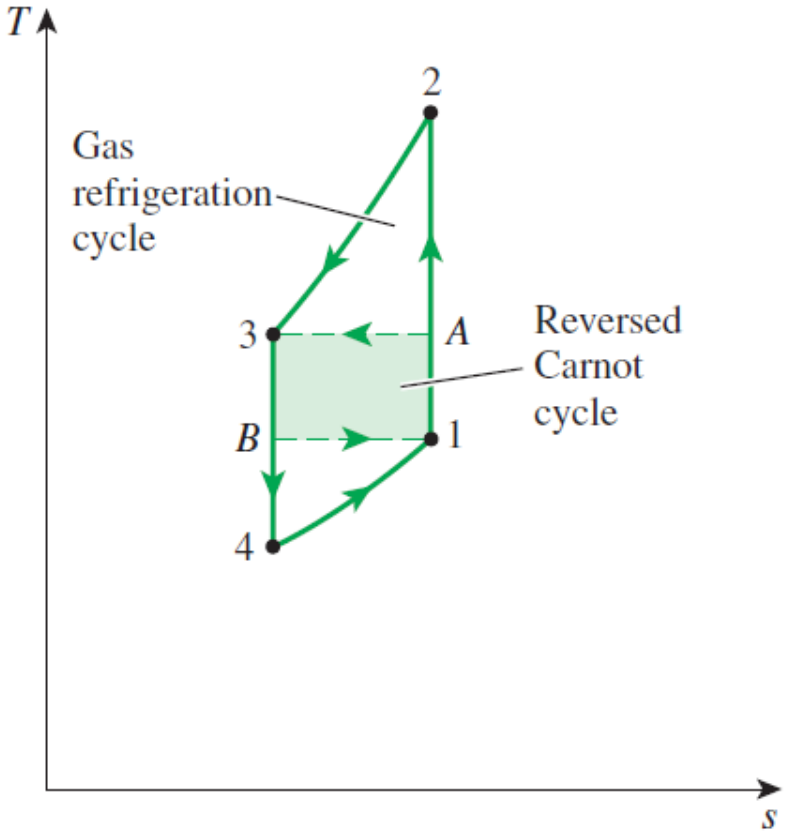


FIGURE 11–19
A reserved Carnot cycle produces more refrigeration (area under *B*1) with less work input (area 1*A*3*B*).

- The reversed Carnot cycle consumes a fraction of the net work (area 1*A*3*B*) but produces a greater amount of refrigeration (triangular area under *B*1).

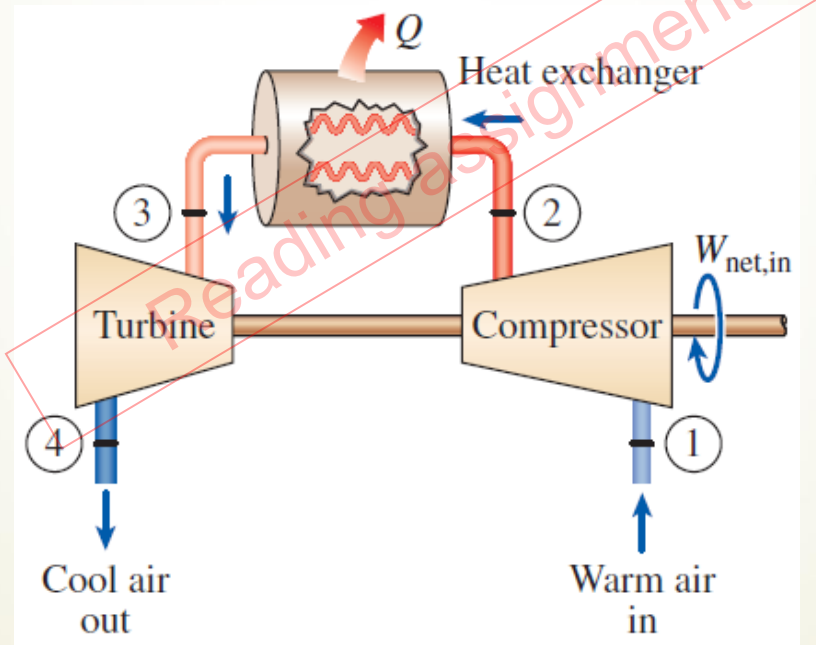


FIGURE 11–20
An open-cycle aircraft cooling system.

Reading assignment

- Despite their relatively low COPs, the gas refrigeration cycles involve simple, lighter components, which make them suitable for aircraft cooling, and they can incorporate regeneration

11.9 Gas Refrigeration Cycles (3)

- Without regeneration, the lowest turbine inlet temperature is T_0, the temperature of the surroundings or any other cooling medium.
- With regeneration, the high-pressure gas is further cooled to T_4 before expanding in the turbine.

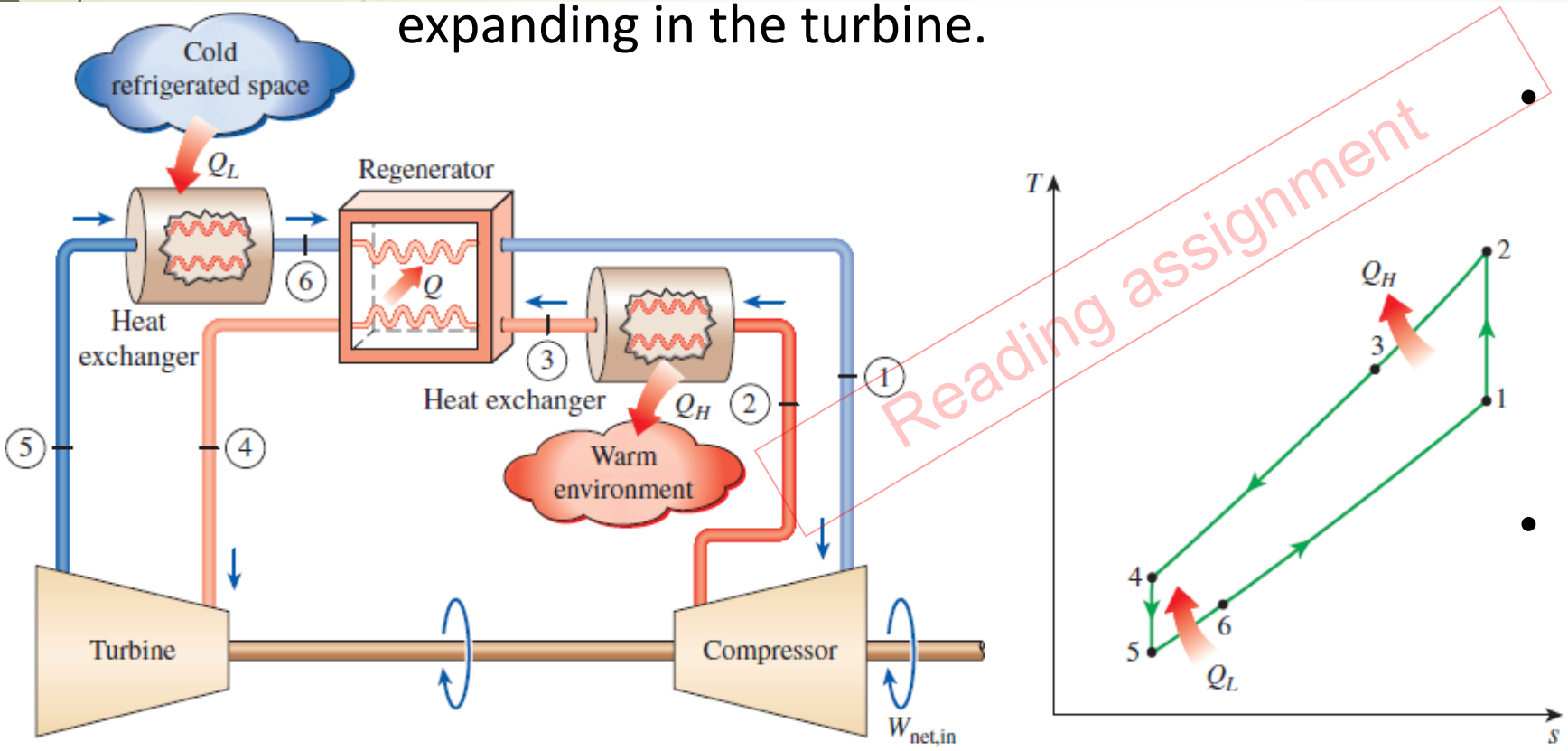


FIGURE 11–21
Gas refrigeration cycle with regeneration.

Reading assignment

- Lowering the turbine inlet temperature automatically lowers the turbine exit temperature, which is the minimum temperature in the cycle.
- Extremely low temperatures can be achieved by repeating regeneration process.

Example 3

EXAMPLE 11–6 The Simple Ideal Gas Refrigeration Cycle

An ideal gas refrigeration cycle using air as the working medium is to maintain a refrigerated space at 0°F while rejecting heat to the surrounding medium at 80°F. The pressure ratio of the compressor is 4. Determine (*a*) the maximum and minimum temperatures in the cycle, (*b*) the coefficient of performance, and (*c*) the rate of refrigeration for a mass flow rate of 0.1 lbm/s.

SOLUTION An ideal gas refrigeration cycle using air as the working fluid is considered. The maximum and minimum temperatures, the COP, and the rate of refrigeration are to be determined.

Assumptions **1** Steady operating conditions exist. **2** Air is an ideal gas with variable specific heats. **3** Kinetic and potential energy changes are negligible.

Analysis The *T-s* diagram of the gas refrigeration cycle is shown in Fig. 11–22. We note that this is an ideal gas-compression refrigeration cycle, and thus, both the compressor and the turbine are isentropic, and the air is cooled to the environment temperature before it enters the turbine.

(*a*) The maximum and minimum temperatures in the cycle are determined from the isentropic relations of ideal gases for the compression and expansion processes. From Table A–17E,

$$T_1 = 460\ \text{R} \longrightarrow h_1 = 109.90\ \text{Btu/lbm} \quad \text{and} \quad P_{r1} = 0.7913$$

Reading assignment

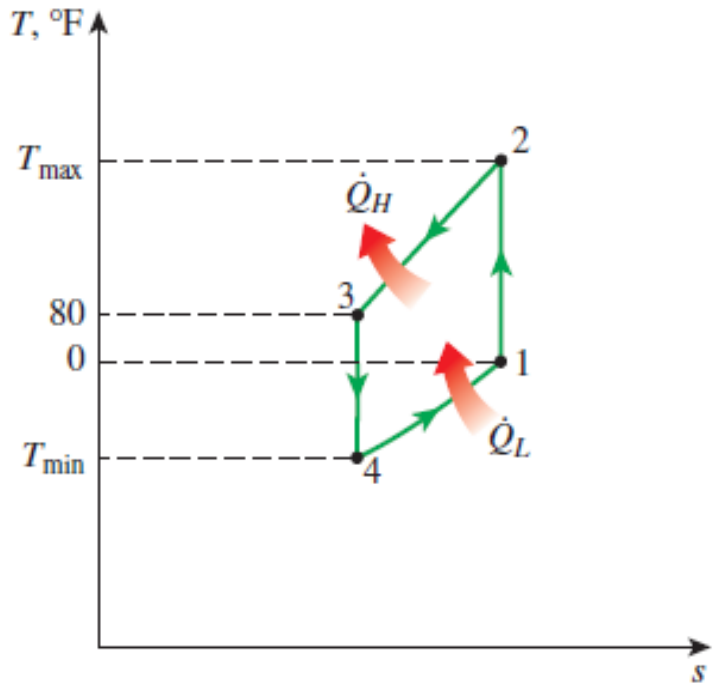


FIGURE 11–22
T-s diagram of the ideal gas refrigeration cycle described in Example 11–6.

$$P_{r2} = \frac{P_2}{P_1} P_{r1} = (4)(0.7913) = 3.165 \longrightarrow \begin{array}{l} h_2 = 163.5 \text{ Btu/lbm} \\ T_2 = \mathbf{683\ R} \text{ (or } \mathbf{223°F}\text{)} \end{array}$$

$$T_3 = 540 \text{ R} \longrightarrow h_3 = 129.06 \text{ Btu/lbm} \quad \text{and} \quad P_{r3} = 1.3860$$

$$P_{r4} = \frac{P_4}{P_3} P_{r3} = (0.25)(1.386) = 0.3456 \longrightarrow \begin{array}{l} h_4 = 86.7 \text{ Btu/lbm} \\ T_4 = \mathbf{363\ R} \text{ (or } \mathbf{-97°F}\text{)} \end{array}$$

Therefore, the highest and the lowest temperatures in the cycle are 223 and −97°F, respectively.

(*b*) The COP of this ideal gas refrigeration cycle is

$$\text{COP}_R = \frac{q_L}{w_{\text{net,in}}} = \frac{q_L}{w_{\text{comp,in}} - w_{\text{turb,out}}}$$

where

$$q_L = h_1 - h_4 = 109.9 - 86.7 = 23.2 \text{ Btu/lbm}$$
$$w_{\text{turb,out}} = h_3 - h_4 = 129.06 - 86.7 = 42.36 \text{ Btu/lbm}$$
$$w_{\text{comp,in}} = h_2 - h_1 = 163.5 - 109.9 = 53.6 \text{ Btu/lbm}$$

Thus,

$$\text{COP}_R = \frac{23.2}{53.6 - 42.36} = \mathbf{2.06}$$

(*c*) The rate of refrigeration is

$$\dot{Q}_{\text{refrig}} = \dot{m} q_L = (0.1 \text{ lbm/s})(23.2 \text{ Btu/lbm}) = \mathbf{2.32\ Btu/s}$$

Discussion It is worth noting that an ideal vapor-compression cycle working under similar conditions would have a COP greater than 3.

Reading assignment

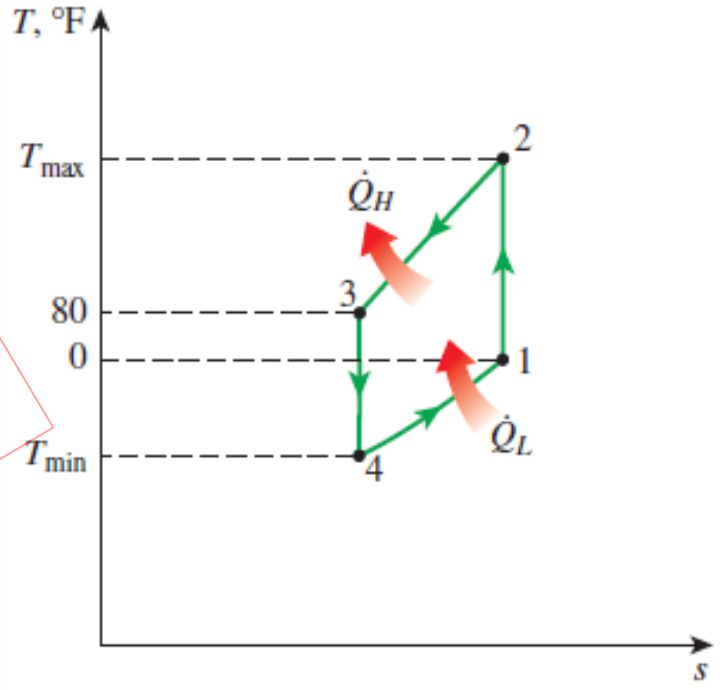


FIGURE 11–22
T-s diagram of the ideal gas refrigeration cycle described in Example 11–6.

Thank You